

A055 — Blind Topological Particle Spectrum Falsifier Chiral Covariance and Directional-Bias Certification

SST-Workbench

Version v0.4.1 — SST Falsifier Framework v1.0.6

1 Identification, scope, and status

A055 v0.4.1 is a new frozen protocol version derived from the v0.4.0 circulation-relative chirality test. The registered equations, thresholds, sealed A054 spatial ladder, and Jacobian finite-difference values are unchanged. v0.4.1 changes execution/provenance semantics: it pins SST Falsifier Framework v1.0.6, imports the A054 isolated runner from an exact manifest allowlist, excludes unlisted executable artifacts from that import namespace, and records computable diagnostics without allowing them to close blocked science gates.

2 Research question and objective

Does a representation-covariant circulation-relative directional bias persist across the frozen numerical controls of the anonymous three-component compound campaign?

3 Competing hypotheses and allowed conclusions

H_0 . No anonymous candidate retains the same circulation-relative directional-bias sign in at least two sectors across all registered resolutions and both Jacobian epsilon values.

H_1 . At least one anonymous candidate does. A PASS supports only a linear finite-model chirality signal; nonlinear confinement and particle identity remain unestablished.

4 Assumptions, domain of validity, and edge cases

The projected A054 Jacobian/mode basis is treated only as a finite-model diagnostic. Curve orientation and circulation sign form a representation pair: reversing both must not change a physical direction observable. v0.3.1 is discovery evidence from the same source family, not independent confirmation. A sector with fewer than three eligible modes cannot claim one-way bias. The A054 runner may be imported only from files named in its runner manifest and matching their registered SHA-256 values; extra executable files are not evidence and are excluded from the import mirror.

5 Symbols, units, and dimensional consistency

Symbol	Meaning	Unit
$P_{jkc}^{\pm}$	component signed spatial-harmonic power	1
$\xi_{j,k}$	circulation-relative directional purity	1
$\eta_{j,k}$	harmonic participation	1
ζ_j	Kelvin-family fraction	1

All decision observables are dimensionless.

6 Exact mathematical model and observables

$$P_{jkc}^{\pm} = |\widehat{a}_{n,c}(\pm k)|^2 + |\widehat{a}_{b,c}(\pm k)|^2. \quad (1)$$

$$\xi_{j,k} = \frac{\sum_c \text{sgn}(\Gamma_c)(P_{jkc}^+ - P_{jkc}^-)}{\sum_c (P_{jkc}^+ + P_{jkc}^-)}. \quad (2)$$

$$\eta_{j,k} = \frac{\sum_c (P_{jkc}^+ + P_{jkc}^-)}{P_{j,\perp}}, \quad B_s = \frac{N_+ - N_-}{N_+ + N_-}. \quad (3)$$

The key covariance identity is

$$\xi[C_c(-s), -\Gamma_c] = \xi[C_c(s), \Gamma_c]. \quad (4)$$

A mode is eligible at its dominant registered harmonic when $\zeta \geq 0.50$, $\eta \geq 0.50$, $|\xi| \geq 0.60$, and $|\Re \lambda|/|\Im \lambda| \leq 1.0$. A sector requires at least three eligible modes and $|B_s| \geq 0.50$.

Machine-readable science contract

Research question After correcting spatial propagation for filament orientation/circulation and enforcing representation covariance, does the one-way traveling-wave imbalance discovered in v0.3.1 persist as a resolution- and Jacobian-epsilon-stable property of anonymous three-component vortex compounds?

Objective Falsify a physical chirality interpretation of the v0.3.1 directional imbalance by replacing raw signed Fourier direction with a circulation-relative gauge-covariant observable, testing metamorphic representation invariance, and requiring recurrence over the sealed A054 spatial-resolution ladder and both preregistered Jacobian finite-difference steps.

H_0 H0: the v0.3.1 one-way imbalance is representation-, resolution-, or finite-difference-sensitive; no anonymous candidate retains a same-sign circulation-relative directional bias in at least two sectors across all frozen numerical controls.

H_1 H1: at least one anonymous candidate retains a same-sign circulation-relative directional bias in at least two sectors under all frozen representation, resolution, and finite-difference controls. This is a finite-model chirality signal only, not particle identification.

Registered equations

E1: Component-resolved signed powers

$$P_{jkc}^{\pm} = |\widehat{a}_{n,c}(\pm k)|^2 + |\widehat{a}_{b,c}(\pm k)|^2 \quad (5)$$

Dimensional check: Squared normalized projected amplitudes are dimensionless.

E2: Circulation-relative directional purity

$$\xi_{j,k} = \frac{\sum_c \text{sgn}(\Gamma_c)(P_{jkc}^+ - P_{jkc}^-)}{\sum_c (P_{jkc}^+ + P_{jkc}^-)} \quad (6)$$

Dimensional check: Ratio of equal-dimension powers; xi is dimensionless.

E3: Harmonic participation

$$\eta_{j,k} = \frac{\sum_c (P_{jkc}^+ + P_{jkc}^-)}{P_{j,\perp}} \quad (7)$$

Dimensional check: Ratio of powers; dimensionless.

E4: Sector directional vote bias

$$B_s = \frac{N_+ - N_-}{N_+ + N_-} \quad (8)$$

Dimensional check: Count ratio; dimensionless.

E5: Representation covariance identity

$$\xi[C_c(-s), -\Gamma_c] = \xi[C_c(s), \Gamma_c] \quad (9)$$

Dimensional check: Equality between dimensionless observables.

E6: Kelvin-family coefficient fraction

$$\zeta_j = \frac{\sum_{i \in \mathcal{K}} |v_{ij}|^2}{\sum_i |v_{ij}|^2} \quad (10)$$

Dimensional check: Ratio of squared normalized coordinates; dimensionless.

Registered code-to-mathematics steps

1. **S1** [G1; equations E5]: Verify sealed A054 source and framework-manifest-valid v0.3.1 discovery output without reading semantic private mappings.
2. **S2** [G2; equations E1]: Recompute projected Jacobians on the sealed spatial ladder and require fine-resolution eigenvalue-multiset agreement with the sealed spectrum.
3. **S3** [G3; equations E1, E2, E3, E4, E6]: For each positive-frequency mode compute component signed powers, Kelvin fraction, dominant harmonic, circulation-relative purity and sector vote bias.
4. **S4** [G2; equations E2, E5]: Apply gauge-equivalent cyclic-shift, rigid-rotation, component-permutation and orientation-reversal-plus-circulation-flip metamorphic controls to selected qualified modes.
5. **S5** [G5; equations E1]: Certify signed-harmonic powers with instance-local C++17/OpenMP FP64 against Python FP64.
6. **S6** [G8; equations E2, E4, E5]: Require same-sign sector bias above threshold at every sealed resolution and at both preregistered Jacobian epsilon values, recurring in at least two sectors of one anonymous candidate.
7. **S7** [G9; equations E4]: After commitment-verified reveal only, attach architecture/component semantics and compute paired contrasts using the frozen blind flags and biases.

7 Exact algorithm: equation-to-code mapping

- S1. Verify A054 and v0.3.1 provenance without semantic reveal.
- S2. Construct an import-only A054 runner mirror from exactly the runner-manifest file set and verify every copied byte by SHA-256; unlisted executables are omitted.
- S3. Recompute projected Jacobians over the sealed resolution ladder and verify the fine sealed spectrum.
- S4. Compute component-resolved signed powers and circulation-relative mode direction.
- S5. Apply cyclic-origin, rigid-rotation, component-permutation, and orientation-reversal plus circulation-flip metamorphic controls.
- S6. Independently certify the signed-power kernel with instance-local C++ FP64 parity, even when an upstream numerical gate is blocked.
- S7. Require same-sign sector bias over all registered resolutions and both Jacobian epsilon values, recurring in at least two sectors. A clean G3 FAIL does not short-circuit G8.
- S8. Reveal semantic labels only after the frozen reveal policy, output-integrity checks and commitments pass.

8 Source, provenance, and independence contract

The authoritative numerical source remains the sealed A054 v0.2.0-r2 compound campaign. A055 v0.3.1 remains a frozen discovery baseline from the same simulation family. Neither is an independent experimental replicate. Missing, ambiguous, partial, or hash-invalid upstream science data is UNRESOLVED. A native loader ambiguity is handled only by reconstructing an import mirror from the exact byte hashes already declared by the upstream runner manifest; the upstream campaign is never modified.

9 Gate DAG and decision semantics

Gate	Question	PASS meaning
G0	Frozen protocol + implementation?	byte-identical
G1	Upstream provenance?	valid
G2	Reference numerics and representation covariance?	spectrum/selftest plus metamorphic error $\leq 10^{-8}$
G3	Fine circulation-relative bias?	at least one sector
G4	Independent confirmation?	disabled
G5	C++ parity?	relative L2 $\leq 10^{-10}$; independent of G2
G6	GPU?	disabled
G7	Independent confirmation?	disabled
G8	Resolution/epsilon recurrence?	at least one anonymous robust candidate; requires G2+G5, not G3 PASS
G9	Reveal integrity?	framework commitments verified

Diagnostic continuation. A055 v0.4.1 binds an instance-local implementation manifest into the frozen science contract and verifies it at G0. It also writes `A055_DIAGNOSTIC_LEDGER.json`. Values computed after a failed or unresolved prerequisite are explicitly diagnostic-only and never alter the official gate ledger. This allows failure structure to be inspected without turning a blocked path into a PASS.

10 Numerical certification, convergence, and error budget

The A054 sealed ladder and Jacobian epsilon values are inherited unchanged. Fine-resolution eigenvalues must reproduce the sealed spectrum within assignment-relative error 10^{-8} . Representation metamorphic error is bounded by 10^{-8} . Directional recurrence uses the same fixed thresholds at every resolution and both epsilon values; no threshold is fit to the v0.3.1 effect size.

11 Backend authority and parity

Python/NumPy FP64 defines the circulation-relative analysis. Instance-local C++17/OpenMP FP64 certifies the underlying signed-harmonic powers against Python with relative $L2 \leq 10^{-10}$. G5 is independent of G2 because it is a deterministic local kernel certification. Production A054 re-computation still requires the backend class recorded by the sealed A054 campaign; no silent fallback can close G2 or G5.

12 Blindness and reveal boundary

Blind computation uses anonymous candidate IDs, sector codes, sealed geometry/numerics, circulation signs, and frozen thresholds only. Semantic architecture/component labels and historical target hypotheses remain private until reveal commitment verification. Framework v1.0.6 additionally verifies blind output-manifest integrity, gate-ledger self-hash, summary-to-ledger binding, frozen protocol identity, unresolved-gate policy, and nonced commitments before reveal.

13 Statistics and multiplicity

G8 is a recurrence/invariance gate, not a population-significance claim. The minimum of three eligible modes prevents a one-mode sector from generating a unit vote bias. Reveal-only architecture contrasts are paired by component assignment, provider stratum and sector when a matched control exists; pooled unpaired medians are not primary evidence.

14 Results insertion contract

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	50e43afed3a71abe03bfa42aa93ba2b9c494da9d0654583baf8548f9741bf954
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	1f0289fc0dac90cad9b0b785bfabb35292c547273aabfc46674b5f1810913976

Gate ledger

Gate	Status	Question
G0	PASS	Are the frozen framework protocol and instance implementation commitment in
G1	PASS	Are the sealed A054 compound source and completed v0.3.1 discovery baseline
G5	PASS	Does strict instance-local C++ FP64 certification reproduce Python signed-har
G4	NOT_APPLICABLE	Is there an independent held-out physical confirmation source?
G6	NOT_APPLICABLE	Is optional GPU screening enabled and parity-qualified?
G7	NOT_APPLICABLE	Is an independent held-out physical confirmation source available?
G2	PASS	Do sealed-spectrum recomputation, analytic covariance selftests, and real-mode
G3	PASS	Does any anonymous sector show the preregistered fine-resolution circulation-r
G8	FAIL	Does any anonymous candidate retain a same-sign circulation-relative direction
G9	PASS	After blind gates, do reveal commitments verify and paired semantic contrasts

Backend manifest

```
{
  "a054_backend": {
    "actual": "cpp_pybind11_openmp",
    "sealed": "cpp_pybind11_openmp"
  },
  "framework_version": "1.0.6",
  "instance_cpp_metric": {
    "actual_backend": "openmp",
    "authority": "CERTIFICATION",
    "build": {
      "actual_backend": "openmp",
      "artifact": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A055_Blind",
      "backend_info": {
        "backend": "openmp",
        "kernel": "biot_savart_plus_signed_harmonic_powers",
        "openmp_compiled": true,
        "openmp_max_threads": 20,
        "precision": "float64"
      },
      "compiler": "C:\\Program Files\\Microsoft Visual Studio\\2022\\Community\\VC\\Tools\\MSVC",
      "compiler_family": "msvc",
      "compiler_version": "Microsoft (R) C/C++ Optimizing Compiler Version 19.44.35228 for x64",
      "error": null,
      "fingerprint_sha256": "e3d7900c880f57b7f08d74a553d440e13d5ba9026fb4ce27f27a2886a6d446e6",
      "flags": [
        "C++17",
        "02/03",
        "OpenMP"
      ],
      "python_abi": "cp314-win_amd64",
      "success": true,
      "toolset": "14.44.35207"
    },
    "parity": {
      "candidate_norm": 139.24626069569086,
      "pass": true,
```

```

    "reference_norm": 139.24626069569092,
    "relative_l2": 6.215789641365356e-16,
    "tolerance": 1e-10
  },
  "status": "PASS"
},
"schema": "A055-BACKEND-MANIFEST-041",
"status": "CHIRALITY_NOT_CERTIFIED",
"summary": {
  "chiral_covariance": {
    "fine_biased_sector_count": 7,
    "max_metamorphic_abs_error": 6.661338147750939e-16,
    "mechanism_promoted_sector_count": 0,
    "robust_candidate_count": 0,
    "row_count": 288
  },
  "diagnostic_only": {
    "analytic_covariance_selftest": {
      "dominant_harmonic": 1,
      "errors": {
        "component_permutation": 0.0,
        "cyclic_shift": 0.0,
        "orientation_reverse_gamma_flip": 0.0,
        "rigid_rotation": 0.0
      },
      "max_abs_error": 0.0,
      "pass": true,
      "values": {
        "canonical": 1.0,
        "component_permutation": 1.0,
        "cyclic_shift": 1.0,
        "orientation_reverse_gamma_flip": 1.0,
        "rigid_rotation": 1.0
      }
    },
    "computed_despite_later_gate_outcomes": {
      "evaluated_sectors": 288,
      "fine_directionally_biased_sectors": 7,
      "max_eigenvalue_assignment_relative_error": 0.0,
      "max_metamorphic_abs_error": 6.661338147750939e-16,
      "mechanism_promoted_sector_count": 0,
      "robust_candidate_count": 0
    },
    "instance_cpp_metric": {
      "actual_backend": "openmp",
      "authority": "CERTIFICATION",
      "build": {
        "actual_backend": "openmp",
        "artifact": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A055",
        "backend_info": {
          "backend": "openmp",
          "kernel": "biot_savart_plus_signed_harmonic_powers",
          "openmp_compiled": true,

```

```

    "openmp_max_threads": 20,
    "precision": "float64"
  },
  "compiler": "C:\\Program Files\\Microsoft Visual Studio\\2022\\Community\\VC\\Tools\\
  "compiler_family": "msvc",
  "compiler_version": "Microsoft (R) C/C++ Optimizing Compiler Version 19.44.35228 for
  "error": null,
  "fingerprint_sha256": "e3d7900c880f57b7f08d74a553d440e13d5ba9026fb4ce27f27a2886a6d440
  "flags": [
    "C++17",
    "O2/O3",
    "OpenMP"
  ],
  "python_abi": "cp314-win_amd64",
  "success": true,
  "toolset": "14.44.35207"
},
"parity": {
  "candidate_norm": 139.24626069569086,
  "pass": true,
  "reference_norm": 139.24626069569092,
  "relative_l2": 6.215789641365356e-16,
  "tolerance": 1e-10
},
"status": "PASS"
},
"mode": "FULL",
"official_gate_semantics_unchanged_by_diagnostics": true,
"schema": "A055-DIAGNOSTIC-LEDGER-041"
},
"discovery_baseline": {
  "guard": "Discovery baseline only; raw signed direction is representation-dependent and i
  "negative_direction_components": 26,
  "positive_direction_components": 238,
  "qualified_traveling_components": 264,
  "schema": "A055-V031-DISCOVERY-BASELINE-040",
  "sectors_with_both_directions": 16,
  "source_output": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A055
},
"framework_version": "1.0.6",
"interpretation": "A G8 PASS is evidence only for a representation-covariant, resolution/ep
"mode": "FULL",
"schema": "A055-BLIND-SUMMARY-041",
"status": "CHIRALITY_NOT_CERTIFIED"
}
}

```

Source manifest

```

{
  "manifest_sha256": "a5cd68e6c5b919cdd41bc845a298bd62ed875afea5dfed7b87c10d118caaf06a",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [

```



```

{
  "catalog_id": "A054",
  "directory_manifest_sha256": "d3e5dc9bbc03bb4e59fe99602381dc2e6797ae4729837c5d46ac0b46777",
  "evidence_class": "simulation",
  "exists": true,
  "file_count": 170,
  "independence_family": "A054_FINITE_CORE_COMPOUND",
  "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A054_nucleon",
  "provenance_family": "A054_VO20_R2_SEALED",
  "registry_hint": {
    "catalog_id": "A054",
    "hits": []
  },
  "role": "primary_reanalysis_source",
  "source_id": "A054_COMPOUND",
  "version": "v0.2.0-r2"
},
{
  "catalog_id": "A055",
  "directory_manifest_sha256": "6beaf1187a1351c5451e15ef6cba6fcc5a1680e6ee47f911a9e264b681",
  "evidence_class": "simulation",
  "exists": true,
  "file_count": 22,
  "independence_family": "A055_VO31_SAME_A054_FAMILY",
  "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A055_Blind_Topological_Particle_Spectrum",
  "provenance_family": "A055_VO31_FRAMEWORK_SEALED",
  "registry_hint": {
    "catalog_id": "A055",
    "hits": []
  },
  "role": "discovery_baseline",
  "source_id": "A055_VO31_DISCOVERY",
  "version": "v0.3.1"
}
],
"workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}

```

15 Allowed interpretation

G3 PASS with G8 FAIL means a finest-grid directional imbalance exists but is not numerically robust. G3 FAIL with valid G2 and G5 is allowed to continue to G8, which should then formally test the registered recurrence criterion rather than remain unexecuted. G2 FAIL from the metamorphic/reference subgate identifies a representation or numerical artifact and blocks chirality interpretation. G8 PASS supports a circulation-relative, representation-covariant, resolution/epsilon-stable linear chirality signal in the finite model only. Upstream spatial-convergence/RPO status is reported separately and remains required before nonlinear-mechanism language is allowed.

16 Reproducibility and exact commands

Canonical instance: 01_research/A_falsifiers/A055_Blind_Topological_Particle_Spectrum/A055-v0.4.1. Canonical framework: 06_templates/SST_Falsifier_Framework/SST_Fal

`sifier_Framework_v1.0.6`. The instance contains a relative `.sst_framework_root` pin. Typical blind production command: `run_blind_only.cmd FULL C:\workspace\projects\SST-Workbench`. Strict reveal remains separate; `run_reveal_if_allowed.cmd` performs only the framework policy preflight/reveal path. Protocol mutation requires a new A055 version.

References

- [1] SST-Workbench, 2026, “A054 v0.2.0-r2 sealed compound campaign”, internal reproducible simulation artifact.
- [2] SST-Workbench, 2026, “A055 v0.3.1 signed-travel discovery output”, internal reproducible simulation artifact.
- [3] SST-Workbench, 2026, “SST Falsifier Framework v1.0.6 CANONICAL_FROZEN”, internal framework release.